AMAN MEHTA

Electrical & Electronics Engineer | Power Systems | Power Electronics & Embedded Systems

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OBJECTIVE

Final-year B.Tech Electrical & Electronics Engineering student (2027 batch) with hands-on exposure to power systems through two vocational trainings at CSPDCL, and project experience in power electronics, power quality and embedded systems (STM32, ESP32, MATLAB/Simulink). Seeking the Graduate Engineer Trainee (Electrical) role at Reliance Industries Limited to build a career in plant electrical engineering, along with working knowledge of software development.

EDUCATION

Chhattisgarh Institute of Technology (CGIT), Raipur *2023 – 2027*
B.Tech — Electrical & Electronics Engineering (Pursuing)

Sarswati Shishu Vidya Mandir, Nawapara (Raipur)
12th – Science (PCM): 87.4% | 10th: 95.3%

INDUSTRY TRAINING

Vocational Training — CSPDCL (Chhattisgarh State Power Distribution Co. Ltd.) *Jul 2026*
Second training term in Electrical Engineering & Power Systems at CSPDCL, following the 2025 training.

Vocational Training — CSPDCL (Chhattisgarh State Power Distribution Co. Ltd.) *Jul 2025*
Electrical Engineering & Power Systems training at CSPDCL.

PROJECTS

Pure Sine Wave Inverter with Power Quality Monitoring
- Designing a transformerless pure sine wave inverter on STM32 — boost converter, unipolar SPWM H-bridge and LC filter — with real-time mains THD sensing for auto charge/battery switchover and overcurrent, overvoltage, thermal and short-circuit protection.

PV-STATCOM — Power Quality Analysis in MATLAB/Simulink
- Simulated a single-phase grid-integrated solar PV-STATCOM (VSI with hysteresis current control, day and night modes); cut grid current THD from 26.45% to 1.34% with unity power factor, meeting IEEE-519.

Waste-Heat Water Distillation for AI Data Centers — ESP32 Hardware Prototype
- Built a working prototype that reuses data center waste heat to distill dirty or salty water; ESP32 switches the heater via relay and monitors a waterproof temperature sensor with automatic overheat cutoff.
- Designed a sealed evaporation chamber with an angled condensing lid and finned pre-warming lid; fixed leaks and weak condensation using silicone gaskets, lid angle and insulation. Published a write-up.

VyomsTech PCB Designer — Browser-Based PCB Design Tool
- Built a PCB design tool with live PCB/schematic/3D preview, real SPICE-based circuit simulation and one-click Gerber export; added Trace, an in-house AI assistant that generates a full circuit from a plain-language description.

SKILLS

Electrical & Power Systems: Power Electronics, Power Quality Analysis (THD, IEEE-519), Grid-Integrated Solar (PV-STATCOM), Power Systems fundamentals (CSPDCL training)
Embedded & Hardware: Embedded C (STM32, ESP32), PCB Design, SPICE Circuit Simulation, Relay and Sensor Interfacing
Simulation & Tools: MATLAB/Simulink, Git/GitHub, VS Code
Additional (Software & CS): Python, C, C++, JavaScript, Node.js, React.js, FastAPI, REST APIs, MongoDB, SQL, DSA, Operating Systems, OOPs, Computer Networks

ADDITIONAL EXPERIENCE (SOFTWARE)

VyomsTech — Full-Stack Development & Work Management Project *2025 – Present*
- Self-directed project where I plan, build and ship web, mobile and AI-integrated products, managing requirements, delivery and client communication on my own.
- Built a responsive Substation Management website (with pinch-zoom single-line diagram) for CSPDCL's Nawapara Rajim 33/11 kV substation, and managed the website, hosting, SEO and email for Ghosh Global Services.
- Built Sangam, an alumni networking platform for CGIT, and GateHub, a web + mobile academic resource platform.

CERTIFICATIONS & ACHIEVEMENTS

- GeeksforGeeks — 160 Days of DSA with Python (Aug 2025) | PhysicsWallah — Decode DSA with Python (Aug 2025)
- Cisco Networking Academy — Data Science Essentials (Dec 2025) | PhysicsWallah — Cloud Computing
- Solved 200+ DSA problems across GeeksforGeeks and LeetCode.